Contribution Journal

In this lab, I independently completed the end-to-end data preparation workflow on the Titanic dataset, addressing data quality issues to prepare the dataset for machine learning models. I resolved missing values in the numerical features by implementing median imputation on the Age column using modern Pandas syntax to avoid deprecation warnings. I then applied one-hot encoding via pd.get_dummies() with drop_first=True to transform categorical attributes (Sex and Embarked) into binary flags without introducing multicollinearity. Finally, I standardized the continuous numerical features (Age and Fare) using Scikit-Learn's StandardScaler to ensure zero mean and unit variance. All code cells were successfully executed, verified for reproducibility, and paired with analytical responses evaluating feature scaling and imputation strategies.